

Re-feeding Is Not Replaying: Transcript Re-feeds Flip the Sign of Counterfactual Token-Credit Estimates

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Abstract

Per-token counterfactual credit estimation is the standard tool for asking which token in a language-model rollout caused the final answer to be right or wrong: cut the transcript at a pivot token, substitute an alternative, replay continuations, and compare outcomes. Every published method we could verify implements the replay by *re-feeding* the transcript prefix through the model as a fresh prompt, implicitly assuming that a re-fed prefix reproduces the state the model actually passed through during generation. We test that assumption directly on a stock inference engine, using a three-pass design: continuations resumed from the exact decode-time KV state, an identical second exact pass that serves as a replica noise floor, and a re-feed pass in which the only changed variable is the prefill path. Across four configurations and two models, including a GRPO-trained checkpoint, re-feeding flips the sign of the credit estimate at decision tokens at rates 20–27 percentage points above the replica floor (paired $z = 2.6$ – 9.3). At the probe level, resuming the decode-time state reproduces the next-token logprob bit-exactly at 78% of pivots; re-feeding does so at 3.9%. Even the floor is informative: batch-composition numerics alone flip greedy outcomes at roughly 14% of low-margin pivots. The experiment was pre-registered with a locked decision rule and cost under \$2 of compute. We recommend that counterfactual credit studies resume exact decoder state (or use batch-invariant kernels) and report a replica floor; without one, a substantial fraction of reported token-level credit signs are attributable to the measurement procedure rather than the model.

1 Introduction

Reinforcement learning with verifiable rewards has made per-token credit assignment a central measurement problem for language-model reasoning. Group relative policy optimization [8] and its descendants train on sequence-level rewards, so a growing line of work asks the finer-grained question: which individual tokens in a rollout actually caused success or failure? The standard answer is counterfactual replay. Choose a pivot token, force an alternative continuation token, sample K completions under each arm, grade them, and read the per-token causal effect off the difference [7, 4, 1].

Every implementation of this recipe that we could verify shares one mechanical step: the transcript prefix up to the pivot is *re-fed* through the model as a fresh prompt, and the continuations branch from the resulting state. The implicit assumption is that re-feeding tokens $x_{1..t}$ reconstructs the internal state the model was in when it originally generated token t . It does not, exactly. Floating-point reduction order inside attention and matrix-multiply kernels depends on batch

composition and on how the sequence is split between prefill and decode, so the re-fed state differs from the original decode-time state in the low-order bits [3]. The working assumption in the credit-assignment literature is that these differences wash out of the final estimate.

This paper measures whether they do. The answer is no: at the low-margin decision tokens that credit methods specifically target, the re-feed shortcut flips the *sign* of the estimated credit at rates far above what measurement noise explains.

Contributions.

1. **A three-pass audit design.** We compare credit estimates from (i) continuations resumed from the exact decode-time KV state, (ii) an identical second exact pass, and (iii) a re-feed pass. Pass (ii) is the load-bearing control: it converts “how often do estimates disagree?” into “how much disagreement is attributable to the prefill path *beyond* replica noise?” The design runs on a stock engine (vLLM [5]) with no kernel modifications.
2. **Quantification.** Pooled over two pre-registered cohorts (40 GSM8K problems, 1,564 pivots), re-feeding flips the credit sign at 33.0% of eligible pivots against a 12.2% replica floor (excess +20.7pp, paired McNemar $z = 9.33$). The effect persists with greedy continuations that eliminate sampling noise entirely (+26.5pp) and on a GRPO-trained checkpoint (+26.3pp).
3. **A bit-level account.** Resuming the decode-time state reproduces the post-pivot next-token logprob *exactly* (delta = 0.0) at 78.0% of 3,127 probes; re-feeding does so at 3.9%, with a 90th-percentile logprob shift of 4.6×10^{-2} .
4. **The floor as a finding.** Even between two bit-identical request batches, continuous-batching numerics flip greedy outcomes at $\sim 14\%$ of low-margin pivots, quantifying a hazard for any single-sample evaluation at decision tokens.

The experiment was pre-registered before any GPU run: temperature, K , grading, the pivot filter, and the paper-versus-null decision rule were locked in advance, and the one amendment (a difficulty screen, after a smoke run showed the model saturates easy problems) is dated and was blind to drift direction. Total compute cost was approximately \$2 on rented A100s. Code, the pre-registration, and every per-pivot record are public.

2 Related Work

Counterfactual token credit. Critical-token fine-tuning [7] forces alternative tokens and regenerates continuations to identify tokens that change outcomes. Khandoga et al. [4] construct causal per-token credit for policy optimization and observe qualitatively that group-relative advantages can carry the wrong sign. Chen et al. [1] show that exact counterfactuals beat approximate ones for credit assignment in cooperative LLM-agent systems, at the level of whole agent actions. All of these estimate counterfactual state by re-feeding or regenerating from text; none resume the original decoder state, and none report a replica control that would bound the contribution of the replay procedure itself. Our result does not diminish these methods; it quantifies a systematic error term that all of them currently absorb silently.

Inference nondeterminism and training mismatch. He and Thinking Machines Lab [3] identify batch invariance as the root cause of LLM inference nondeterminism and fix it with batch-invariant kernels, demonstrating bitwise-deterministic sampling and true on-policy RL. Their victim

is the serving endpoint and the trainer-versus-sampler gap in the RL loop. Zhong et al. [10] diagnose training-inference mismatch, where the rollout engine and the trainer assign different logprobs to the same tokens under the same weights, as an independent cause of RL training collapse. Both lines concern the *training pipeline*. Our victim is different: the *measurement methodology* of the credit-assignment literature, where the comparison pair is not trainer-versus-sampler but re-feed prefill state versus original decode-time state, on the stock engines that methodology actually runs on. The three lines compose: batch-invariant kernels are the fix, mismatch diagnosis covers training, and this paper covers measurement. In a fully batch-invariant engine where a token’s reduction order is independent of its prefill/decode split, the gap we measure should collapse to the replica floor by construction; that is a testable prediction of He and Thinking Machines Lab [3]’s kernels which our harness can evaluate directly.

3 Method

3.1 Setup and estimator

For each problem we sample $N = 8$ trunk rollouts at temperature 0.7 with top- k logprobs recorded. Because the engine caches KV blocks as it decodes, the exact state every rollout passed through remains resident in the prefix cache at block granularity. A *pivot* is a generated token where the top-1/top-2 logprob gap is below 1.0 (a low-margin decision token, the population credit methods target), restricted to positions within 2 tokens past a 16-token block boundary so that the exact arm resumes from cached state with at most 2 recomputed tokens. Per rollout we take at most 5 pivots, earliest first.

At pivot position t with sampled token a and best alternative a' , the credit estimate under a given replay method is

$$\hat{A}_t = \frac{1}{K} \sum_{k=1}^K R(x_{1..t-1}, a, c_k) - \frac{1}{K} \sum_{k=1}^K R(x_{1..t-1}, a', c'_k),$$

where c_k, c'_k are sampled continuations and R grades the completed solution by exact match on the GSM8K [2] final answer. Continuation seeds are a deterministic hash of (problem, rollout, pivot, arm, k), so all replay methods receive identical sampling randomness; the only uncontrolled difference between methods is the prefill path.

3.2 Three passes

Exact pass 1. All continuation requests for a problem are submitted while the trunk’s decode-time KV blocks are resident. The prefix cache-hits the state the rollout actually traversed; continuations resume from it.

Exact pass 2 (replica floor). The identical request batch is submitted again, cache still warm. Any disagreement between passes 1 and 2 is attributable to continuous-batching numerics and sampling-on-perturbed- logits alone: it is the noise floor of the measurement itself.

Re-feed pass. The prefix cache is reset and the identical batch is submitted a third time. The trunk is now freshly prefilled, exactly as in published credit pipelines. Differences beyond the replica floor are attributable to the re-feed prefill path.

The primary metric is the fraction of eligible pivots (those where at least one pass yields $\hat{A}_t \neq 0$) at which $\text{sign}(\hat{A}_t)$ under re-feed disagrees with exact pass 1, reported against the same quantity for

exact pass 2. Each pass also issues one greedy 32-token probe per arm with top-2 logprobs recorded, giving a grading-free channel: probe-sequence divergence rates and the absolute delta of the first post-pivot token’s logprob.

3.3 Why decode-time resume is the right reference

The decode-time KV state is not an alternative approximation; it is the state the model was actually in when it chose token t . A counterfactual *do*-operation at t is defined relative to that state. Resuming it requires no kernel modifications: paged-attention engines already retain it in the prefix cache, and durable snapshot tools can persist it across processes. The replica pass empirically validates the reference: 78% of resumed probes reproduce the next-token logprob bit-exactly (Section 5), whereas a re-fed prefix essentially never does. One honest caveat: blocks are 16 tokens, so “exact” means exact to the block boundary, with at most 2 trailing tokens recomputed under our pivot filter.

3.4 Pre-registration

Temperature, K , the grading function, the pivot filter, and the decision rule (sign-flip excess above 5%: the method is materially unsound; below 2%: re-feed is adequate) were locked before any GPU run. A smoke run revealed that Qwen2.5-7B-Instruct solves easy GSM8K problems in every continuation, making $\hat{A}_t = 0$ everywhere and the metric undecidable for lack of eligible pivots. Amendment 1, dated before any full-run results, added a second cohort screened only on *trunk* rollout accuracy being mixed (neither 0/8 nor 8/8), a criterion computed before any arm continuations exist and blind to drift direction. Both cohorts are reported separately and pooled.

4 Experimental Setup

Models. Qwen2.5-7B-Instruct [6] and Qwen-2.5-7B-SimpleRL-Zoo [9], a GRPO-trained checkpoint built from the Qwen2.5-7B base on mathematical reasoning. **Engine.** Stock vLLM 0.22.1 [5], bfloat16, prefix caching enabled, no kernel modifications. **Hardware.** One A100 SXM 80GB per run (rented; two pods, roughly 80 minutes total, ~\$2). **Data.** GSM8K test split [2]: the first 20 problems (primary cohort) and the first 20 mixed-accuracy problems under the Amendment 1 screen (screened cohort). **Configurations.** Main: temperature 0.7, $K = 4$. Secondary: greedy continuations (temperature 0, $K = 1$), $K = 8$, and the GRPO checkpoint, all on screened cohorts.

5 Results

5.1 Primary metric: sign flips above the replica floor

Table 1 reports, on the common eligible set of each run (pivots where any pass gives $\hat{A}_t \neq 0$, identical denominator for both comparisons), the rate at which re-feed disagrees in sign with exact pass 1, against the exact-versus-exact replica floor, with a paired McNemar test on the discordant pivots.

Every configuration is positive and significant. The two strongest are the two that answer the obvious objections. With greedy continuations there is no sampling randomness at all, and re-feed still flips 40.2% of estimates against a 13.8% pure-numerics floor; per problem, 11 of 19 problems show re-feed worse than floor and zero show the reverse. On the GRPO-trained checkpoint, the population for which credit assignment is actually studied, the excess is +26.3pp and the per-problem split is 16 worse, 2 better. In the pooled main configuration the discordant counts are 160 pivots

Configuration	Model	n	Re-feed flip	Floor	Excess	z
Main, pooled ($t=0.7$, $K=4$)	Qwen2.5-7B-Instruct	622	33.0%	12.2%	+20.7pp	9.33
Greedy ($t=0$, $K=1$)	Qwen2.5-7B-Instruct	189	40.2%	13.8%	+26.5pp	5.66
$K=8$ ($t=0.7$) [†]	Qwen2.5-7B-Instruct	104	35.6%	22.1%	+13.5pp	2.56
GRPO ckpt ($t=0.7$, $K=4$)	SimpleRL-Zoo-7B	373	35.4%	9.1%	+26.3pp	8.66

Table 1: Sign-flip rate of the credit estimate under re-feed versus the exact-replica noise floor, on each run’s common eligible set. z is the paired McNemar statistic over discordant pivots. [†]Truncated at 4/20 problems by a transient pod disk failure unrelated to the harness; reported for completeness.

flipped only by re-feed against 31 flipped only by the replica ($z = 9.33$). The effect is flat in pivot margin: 32.6% below a 0.3 logprob gap versus 33.2% above it (floor 12.2% in both buckets), so this is not confined to near-ties.

5.2 Bit-level account

The greedy probes give a grading-free view of the same phenomenon. Across 3,127 probes pooled over the full runs, the replica pass reproduces the first post-pivot token’s logprob *exactly* (absolute delta = 0.0) at 78.0% of pivots; the re-feed pass does so at 3.9%. When nonzero, the median replica delta and median re-feed delta are both $\sim 1.7 \times 10^{-3}$, but the tails differ by an order of magnitude: the 90th-percentile delta is 2.5×10^{-3} for the replica versus 4.6×10^{-2} for re-feed (maximum 0.18). Probe sequences diverge within 32 greedy tokens at 23.1% of pivots under re-feed versus 6.7% for the replica on the main runs, and at 16.0% versus 1.7% (a $9.4\times$ ratio) on the GRPO checkpoint.

5.3 The floor is also a finding

Two bit-identical request batches, submitted seconds apart against the same warm cache with the same seeds, disagree on the sign of $\hat{A}_t$ at 9–14% of eligible pivots (22% under $K=8$ ’s partial run) and on greedy continuations at 6–9% of pivots. This is He and Thinking Machines Lab [3]’s batch-composition nondeterminism expressed in the unit that matters for evaluation: at the decision tokens where models are uncertain, batching numerics alone flip single-sample outcomes at a rate that would be catastrophic to interpret as signal. Any study that evaluates at low-margin tokens without a replica control is reading a coin flip at roughly one pivot in ten.

6 Discussion

What this does and does not show. It shows that the re-feed shortcut is materially unsound as a measurement procedure for per-token counterfactual credit at decision tokens on stock engines: roughly one in five sign judgments beyond noise is an artifact of the replay path. It does not show that GRPO training is broken (we measured the estimator, not training outcomes), and it does not claim novelty for the existence of batch-composition drift, which He and Thinking Machines Lab [3] established and fixed at the kernel level. The contribution is the quantification, on the measurement methodology the credit literature actually uses, with a control that no prior credit study reports.

Recommendations. (1) Resume the decode-time state for counterfactual replay. On paged-attention engines this is engine-native via the prefix cache within a process, and durable session-snapshot tooling extends it across processes. (2) Alternatively, run on batch-invariant kernels [3],

under which prefill/decode-split invariance should collapse the re-feed gap to the floor; confirming that prediction with this harness is immediate future work. (3) In all cases, report an exact-replica floor. A credit-assignment result at low-margin tokens without a floor is uninterpretable, since the floor alone is $\sim 10\%$. (4) Pre-register estimator settings: the flip rate is sensitive to temperature, K , grading, and the pivot filter, and is easy to tune post hoc.

Limitations. One task family (GSM8K), one engine version (vLLM 0.22.1), one GPU type (A100); pivots are deliberately conditioned on low margin and on block-boundary proximity, so the rates are statements about decision tokens, not about all tokens; K is small (the binary-graded estimator is noisy, which the floor makes explicit); the $K=8$ run is a 4-problem partial; and “exact” is exact to the 16-token block boundary. None of these caveats favor the effect: the strongest results occur exactly where the caveats are weakest (greedy continuations, the GRPO checkpoint).

7 Reproducibility

The pre-registration (with its dated amendment), the harness (a single ~ 500 -line Python file against stock vLLM), every per-pivot record of all five runs, and the run logs are public in the thaw repository (https://github.com/thaw-ai/thaw_benchmarks/). Total compute: two rented A100 pods, roughly 80 minutes and \$2. Each headline number in this paper can be recomputed from the published JSONs with the included analysis function.

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